

- 1** Mr Brown gives his class a test.
The 10 girls in the class get a mean mark of 70%
The 15 boys in the class get a mean mark of 80%

Nick says that because the mean of 70 and 80 is 75 then the mean mark for the whole class in the test is 75%

Nick is not correct.

Is the correct mean mark less than or greater than 75%?

You must justify your answer.

.....

.....

.....

(Total for Question 1 is 2 marks)

2 The mean length of 5 sticks is 4.2 cm.

Nawal measured the length of one of the sticks as 7 cm.

(a) Work out the mean length of the other 4 sticks.

..... cm
(3)

Nawal made a mistake.

The stick was not 7 cm long.

It was 17 cm long.

(b) How does this affect your answer to part (a)?

.....
.....
.....
(1)

(Total for Question 2 is 4 marks)

3 The mean length of 5 sticks is 4.2 cm.

Nawal measured the length of one of the sticks as 7 cm.

(a) Work out the mean length of the other 4 sticks.

..... cm
(3)

Nawal made a mistake.
The stick was not 7 cm long.
It was 17 cm long.

(b) How does this affect your answer to part (a)?

.....
.....
.....
(1)

(Total for Question 3 is 4 marks)

4 There are 30 women and 20 men at a gym.

The mean height of all 50 people is 167.6 cm

The mean height of the 20 men is 182 cm

Work out the mean height of the 30 women.

..... cm

(Total for Question 8 is 3 marks)

- 5 There are 10 boys and 20 girls in a class.
The class has a test.

The mean mark for all the class is 60

The mean mark for the girls is 54

Work out the mean mark for the boys.

.....
(Total for Question 7 is 3 marks)

- 6 4 red bricks have a mean weight of 5 kg.
5 blue bricks have a mean weight of 9 kg.
1 green brick has a weight of 6 kg.

Donna says,

“The mean weight of the 10 bricks is less than 7 kg.”

Is Donna correct?

You must show how you get your answer.

(Total for Question 6 is 3 marks)

7 There is a total of 45 boys and girls in a choir.

The mean age of the 18 boys is 16.2 years.

The mean age of the 27 girls is 16.7 years.

Calculate the mean age of all 45 boys and girls.

..... years

(Total for Question 7 is 3 marks)

8 Len has 8 parcels.

The mean weight of the 8 parcels is 2.5 kg.

The mean weight of 3 of the parcels is 2 kg.

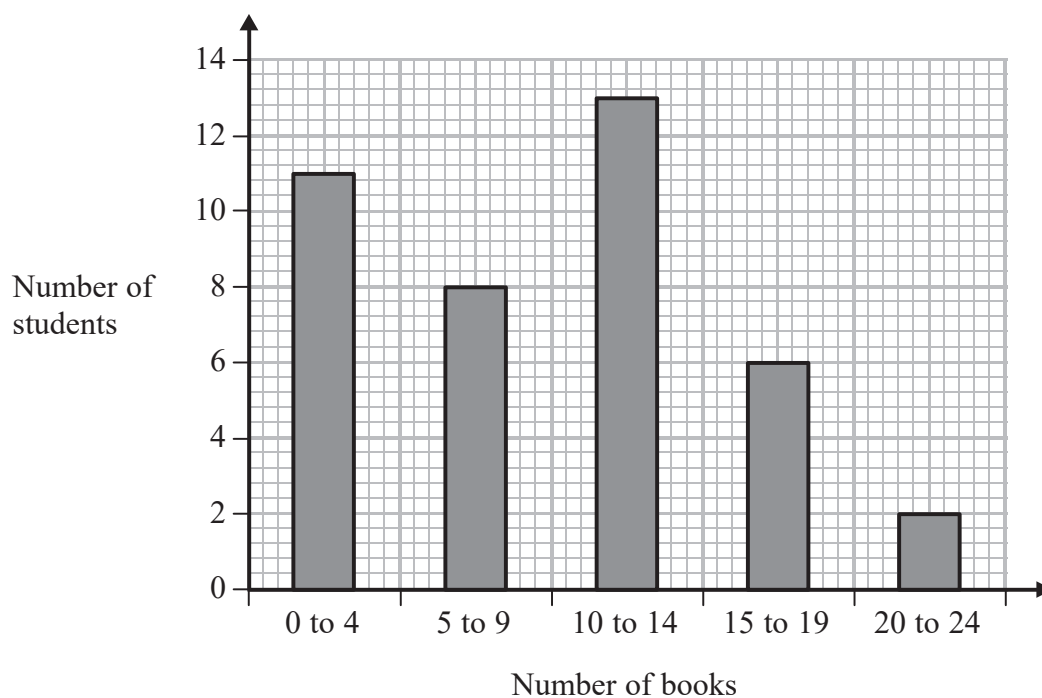
Work out the mean weight of the other 5 parcels.

..... kg

(Total for Question 8 is 3 marks)

- 9 Fran asks each of 40 students how many books they bought last year.

The chart below shows information about the number of books bought by each of the 40 students.



- (a) Work out the percentage of these students who bought 20 or more books.

..... %
(2)

- (b) Show that an estimate for the mean number of books bought is 9.5
You must show all your working.

(4)

(Total for Question 9 is 6 marks)